— *Solution notes* —

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 4

1 Compiler Construction (tgg22)

This question concerns constructing $LL(1)$ parsers from context-free grammars.

Context-free grammars

(*a*) Suppose we have a grammar that contains these productions.

$$\begin{array}{lcl} S & \rightarrow & \text{if } E \text{ then } S \text{ else } S \text{ end} \\ S & \rightarrow & \text{if } E \text{ then } S \text{ end} \end{array}$$

Note that these productions have a shared initial segment. Explain how this prevents us from automatically generating an $LL(1)$ parser directly from this grammar. [2 marks]

Answer: Both productions start with the keyword "if" and so a $LL(1)$ will not be able to select the correct production based on one token look-ahead.

Context-free grammars

(*b*) Rewrite the productions from Part (*a*) so that they could be suitable as input to an $LL(1)$ parser generator. [4 marks]

Answer: We have to have a unique production starting with the keyword "if". One way to do this is to have that production contain the shared initial segment, followed by a non-terminal that can produce either of the final segments.

$$\begin{array}{lcl} S & \rightarrow & \text{if } E \text{ then } S\ F \\ F & \rightarrow & \text{end} \\ F & \rightarrow & \text{else } S \text{ end} \end{array}$$

Here F must be a fresh non-terminal.

Context-free grammars

(*c*) Eliminate the shared initial segments from these grammar productions.

$$\begin{array}{lcl} A & \rightarrow & aAbCeDg \\ A & \rightarrow & aAbCd \\ B & \rightarrow & eDgEf \end{array}$$

[5 marks]

Answer: First, eliminate the shared initial segment from the first two productions.

$$\begin{array}{lcl} A & \rightarrow & aAbCF \\ F & \rightarrow & eDg \\ F & \rightarrow & d \\ B & \rightarrow & eDgEf \end{array}$$

This results in a shared initial segment for the second and last productions. So, we apply the transformation again.

$$\begin{array}{lcl} A & \rightarrow & aAbCF \\ F & \rightarrow & eDgG \\ F & \rightarrow & d \\ G & \rightarrow & Ef \\ G & \rightarrow & \end{array}$$

Context-free grammars

(*d*) Give a general method for eliminating shared initial segments from a grammar. [5 marks]

Answer:

- (1) If there are two productions $A_1 \to \alpha\beta_1$ and $A_2 \to \alpha\beta_2$ having a shared initial segment α, then continue to step (2), otherwise exit.
- (2) Generate a new non-terminal B and replace the two productions above with the set

$$\{A_1 \to \alpha B,\ A_2 \to \alpha B,\ B \to \beta_1,\ B \to \beta_2\}.$$

  Note that it might be that $A_1 = A_2$.
- (3) Go to step (1).

Context-free grammars

(*e*) Argue carefully that the grammar produced by your method generates the same language as the original grammar. [4 marks]

Answer:

Suppose that productions $A_1 \to \alpha\beta_1$ and $A_2 \to \alpha\beta_2$ have been replaced by the set

$$\{A_1 \to \alpha B,\ A_2 \to \alpha B,\ B \to \beta_1,\ B \to \beta_2\}.$$

Suppose that

$$S \to^+ \gamma A_i \alpha\beta_i \lambda \to^+ w$$

is a derivation in the original grammar. Then, the transformed grammar can generate w as

$$S \to^+ \gamma A_i \alpha B \lambda \to \gamma A_i \alpha\beta_i \lambda \to^+ w$$

Conversely, suppose that

$$S \to^+ w$$

is a derivation in the new grammar. Note that productions for B can only be used after a product of either A_1 or A_2 (since B is new and only introduced by the A_i productions).

$$S \to^+ \gamma A_i \alpha B \lambda \to^+ \gamma' B \lambda' \to \gamma' \beta_i \lambda' \to^+ w$$

This derivation of w can be performed in the original grammar as

$$S \to^+ \gamma A_i \alpha\beta_i \lambda \to^+ \gamma' \beta_i \lambda' \to^+ w.$$